Math **163** Name (PLEASE PRINT) Cap'n Orangebeard

Exam **1B (Online)** **2020 Fall**/Spring/Summer I /Summer II Score_____

Please fill in the following table checking boxes corresponding to parts of problems that you have completed and submitted and an expected grade. You will lose double the points that a part is worth if you submit incomplete, unproved, or unchecked answers, as per the directions of the problem. Ideally you should know exactly what you have done correctly and what grade you should receive. I will fill in the received grade after I correct the exam. Not all questions have the same number of parts so that some columns may be redundant for particular problems.

										Expected	Received
	Part	Part	Part	Part	Part	Part	Part	Part	Part	Grade	Grade
	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)	(ix)		
Number											
1a	1	1	1	5						8	
1b											
1c											
2a	1	1	1	1	1	1				6	
2b	3	3								6	
2c											
3a	3	3								6	
3b	2	3								5	
3c											
								Total		31	

Your J-number $JN := \underline{13044034} := (J_k | k \in 1..8)$

Hence

$J_1 = 1$ $J_2 = 3$ $J_3 = 0$ $J_4 = 4$

$J_5 = 4$ $J_6 = 0$ $J_7 = 3$ $J_8 = 4$

R For each of sets defined below **write the elements in increasing order**

D Define $\mathcal{U} := 0..9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

D Define $J := \{J_k \in 0..9 | k \in 1..8\} = \{0, 1, 3, 4\}$

D Define $\forall n \in \mathbb{N}$ $Prm(n) :\Leftrightarrow$ *n is a prime number*

D Define $E := \left\{ J_k \in 0..9 \middle| \cancel{Prm}(k) \right\} = \{0, 1, 4\}$

k	1	2	3	4	5	6	7	8
J_k	(1)	3	0	(4)	4	(0)	3	(4)
$\cancel{Prm}(k)$	(Y)	Y	Y	(Y)	Y	(Y)	Y	(Y)
	N	(N)	(N)	N	(N)	N	(N)	N

D $s := mnm_{k\in 1..8}\{J_k \in 0..9 | J_k \neq 0\} = 1$

D $b := mxm_{k\in 1..8}\{J_k \in 0..9 | k \in 1..8\} = 4$

D Define

$$A := \{0, s, b\} = \{0, 1, 4\}$$

D Define

$$B := \left\{ J_k \in 0..9 \middle| Prm(k) \right\} = \{0, 3, 4\}$$

k	1	2	3	4	5	6	7	8
J_k	1	3	0	4	4	0	3	4
$Prm(k)$	Y	(Y)	(Y)	Y	(Y)	Y	(Y)	Y
	(N)	N	N	(N)	N	(N)	N	(N)

D A partial order on a set is said to be total if and only if every pair of elements is comparable under the ordering. This word appears in 2c and part of the test is to determine if you can formulate this definition precisely using proper symbols and quantification and then use this definition in 2c.

Your J-number $JN :=$ 0 0 3 4 0 1 4 4

We define $\forall k \in 1..8$,

D $J_k :=$ the k[th] entry in your J-number counting from the left.

D $r_k :=$ the remainder obtained upon dividing J_k by 5.

Therefore,

$J_1 := 1 = 0(5) + 1$

$J_2 := 3 = 0(5) + 3$

$J_3 := 0 = 0(5) + 0$

$J_4 := 4 = 0(5) + 4$

$J_5 := 4 = 0(5) + 4$

$J_6 := 0 = 0(5) + 0$

$J_7 := 3 = 0(5) + 3$

$J_8 := 4 = 0(5) + 4$

Show all long divisions on the facing page

$r_1 := 1$

$r_2 := 3$

$r_3 := 0$

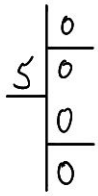
$r_4 := 4$

$r_5 := 4$

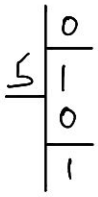
$r_6 := 0$

$r_7 := 3$

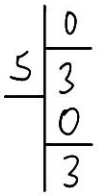
$r_8 := 4$



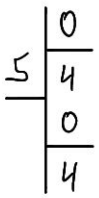
The remainder upon dividing $T_1 = 0$ by 5 is 0; hence $r_1 = 0$



The remainder upon dividing $T_6 = 1$ by 5 is 1; hence $r_6 = 1$



The remainder upon dividing $T_3 = 3$ by 5 is 3; hence $r_3 = 3$



The remainder upon dividing $T_4 = 4$ by 5 is 4; hence $r_4 = 4$

D Define a relation $R \subseteq \{0,1,2,3,4\} \times \{0,1,2,3,4\} =: \mathcal{U}$ as follows:

Compute *R* on the facing side, and write the answer below:

$$R$$

$$:= \{(0,0)\} \cup \left(\bigcup_{k \in 1..4} \{(r_{1+k}, r_{2+k})\} \right)$$

$$= \{(0,0), (0,1), (0,3), (3,4), (4,0)\}$$

R **In order to finish this exam you need to learn by yourself how to multiply matrices**. Schaum's Outlines and the Internet contain information on matrices and may be consulted.

You will need the following definitions.

D Given the matrix for a relation $M(R)$ we define:

$$(M(R))(0,1)$$

$$:= \quad Matrix\ obtained\ by\ replacing\ all\ non-zero\ entries\ in\ M(R)\ by\ 1$$

R **Note that in a set an element may occur exactly once**. Therefore, there may not be any repeated elements in either A or B in the above. **The sets *A* and *B* will occur in several problems on this test. So, if you do not have the correct sets, all your answers will be wrong.**

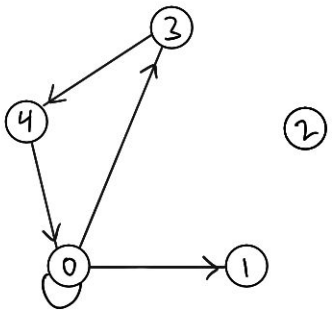
R **In arrow-diagrams, put the numbers in rectangles rather than ovals and draw the rectangles using rulers.**

R **Any sloppiness, untidiness, and any departure from proper format (as indicated in class) will lead to a score of 0**.

$$
\begin{aligned}
&R \\
&:= \{(0,0)\} \cup \left(\bigcup_{k \in 1..4} \{(r_{1+k}, r_{2+k})\} \right) \\
&= \{(0,0)\} \cup \{(r_{1+1}, r_{2+1}), (r_{1+2}, r_{2+2}), (r_{1+3}, r_{2+3}), (r_{1+4}, r_{2+4})\} \\
&= \{(0,0)\} \cup \{(r_2, r_3), (r_3, r_4), (r_4, r_5), (r_5, r_6)\} \\
&= \{(0,0)\} \cup \{(0,3), (3,4), (4,0), (0,1)\} \\
&= \{(0,0), (0,1), (0,3), (3,4), (4,0)\}
\end{aligned}
$$

17 (1abc) Carry out the following instructions, **doing all computations on the facing side** and **recording only the answers on this side**.

1 (i) **Draw directed graph for the relation *R*** earlier **in the space below** with 0, 1, 2, 3, 4 as nodes **arranged in the shape of a regular pentagon with a horizontal base**, with the left vertex of the horizontal side labelled 0, and labelling all the vertices in order traversing them in the counter-clockwise sense.



1 (ii) Find the 5× 5 matrix $M(R)$ for the relation R.

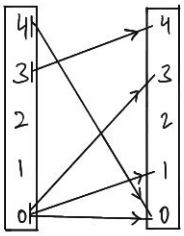
$$M(R) = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1 (iii) Find the 5× 5 matrix $M(R^{op})$ for the relation R^{op}.

$$M(R^{op}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

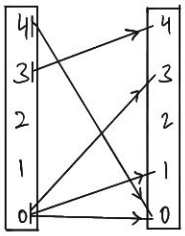
abc. ii. 0

AD(R)

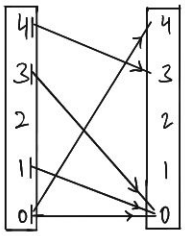


1abc. iii. 0

AD(R)



AD(R^{op})



1abc. ii. 1

M(R)

	0	1	2	3	4
0	1	1	0	1	0
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	0	1
4	1	0	0	0	0

$$M(R) = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1abc. iii. 1

M(R^{op})

	0	1	2	3	4
0	1	0	0	0	1
1	1	0	0	0	0
2	0	0	0	0	0
3	1	0	0	0	0
4	0	0	0	1	0

$$M(R^{op}) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

17 (1abc) Carry out the following instructions

5 (iv) Compute: on the **facing side**:

$$R \circ R^{op}, \qquad R^{op} \circ R,$$

$$M(R \circ R^{op}), \qquad M(R^{op} \circ R),$$

$$\big(M(R)\big)\big(M(R^{op})\big), \qquad \big(M(R^{op})\big)\big(M(R)\big)$$

Prove on **this side** that:

(1) $$\Big(M(R \circ R^{op})\Big)(0,1) = \Big(\Big(M(R^{op})\Big)\Big(M(R)\Big)\Big)(0,1)$$

$$LS = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} = RS$$

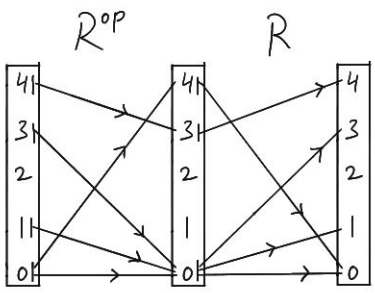
$RS = LS$

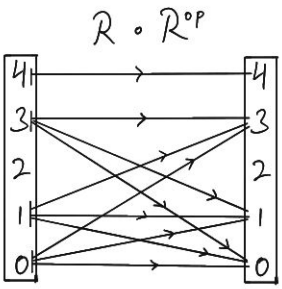
(2) $$\Big(M(R^{op} \circ R)\Big)(0,1) = \Big(\Big(M(R)\Big)\Big(M(R^{op})\Big)\Big)(0,1)$$

$$LS = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} = RS$$

$LS = RS$

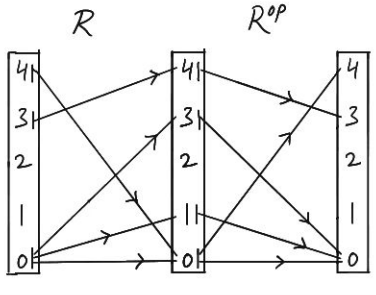
1abc.iv.0

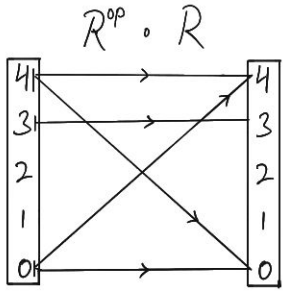




$$\mathcal{R} \circ \mathcal{R}^{op} = \left\{ \begin{matrix} (0,0), (0,1), (0,3), (1,0), (1,1) \\ (1,3), (3,0), (3,1), (3,3), (4,4) \end{matrix} \right\}$$

1abc.iv.1





$$\mathcal{R}^{op} \circ R = \{(0,0), (0,4), (3,3), (4,0), (4,4)\}$$

1abc.iv.2

	0	1	2	3	4
0	1	1	0	1	0
1	1	1	0	1	0
2	0	0	0	0	0
3	1	1	0	1	0
4	0	0	0	0	1

$$M(R \circ R^{op}) = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1abc.iv.3

	0	1	2	3	4
0	1	0	0	0	1
1	0	0	0	0	0
2	0	0	0	0	0
3	0	0	0	1	0
4	1	0	0	0	1

$$M(R^{op} \circ R) = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1abc.iv.4

$$((M(R))(M(R^{op}))) \pmod 2$$

$$= \left(\left(\begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix} \right) \left(\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \right) \right) \pmod 2$$

$$= \begin{bmatrix} a & b & c & d & e \\ f & g & h & i & j \\ k & l & m & n & o \\ p & q & r & s & t \\ u & v & w & x & y \end{bmatrix} \pmod 2$$

$$= \begin{bmatrix} 3 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix} \pmod 2$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1abc.iv.5

$a := (1)(1) + (1)(1) + (0)(0) + (1)(1) + (0)(0) = 3$
$b := (1)(0) + (1)(0) + (0)(0) + (1)(0) + (0)(0) = 0$
$c := (1)(0) + (1)(0) + (0)(0) + (1)(0) + (0)(0) = 0$
$d := (1)(0) + (1)(0) + (0)(0) + (1)(0) + (0)(1) = 0$
$e := (1)(1) + (1)(0) + (0)(0) + (1)(0) + (0)(0) = 1$
$f := (0)(1) + (0)(1) + (0)(0) + (0)(1) + (0)(0) = 0$
$g := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$h := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$i := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(1) = 0$
$j := (0)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$k := (0)(1) + (0)(1) + (0)(0) + (0)(1) + (0)(0) = 0$
$l := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$m := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$n := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(1) = 0$
$o := (0)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$p := (0)(1) + (0)(1) + (0)(0) + (0)(1) + (1)(0) = 0$
$q := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (1)(0) = 0$
$r := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (1)(0) = 0$
$s := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (1)(1) = 1$
$t := (0)(1) + (0)(0) + (0)(0) + (0)(0) + (1)(0) = 0$
$u := (1)(1) + (0)(1) + (0)(0) + (0)(1) + (0)(0) = 1$
$v := (1)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$w := (1)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$x := (1)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(1) = 0$
$y := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 1$

1abl. iv. 6

$$((M(R^{op}))(M(R)))\ (\text{mod } 2)$$

$$= \left(\left(\begin{bmatrix} 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}\right)\left(\begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}\right) \quad (\text{mod } 2)$$

$$= \begin{bmatrix} a & b & c & d & e \\ f & g & h & i & j \\ k & l & m & n & o \\ p & q & r & s & t \\ u & v & w & x & y \end{bmatrix} \quad (\text{mod } 2)$$

$$= \begin{bmatrix} 2 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (\text{mod } 2)$$

$$= \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

1a. iv. 7

$a := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (1)(1) = 2$
$b := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (1)(0) = 1$
$c := (1)(0) + (0)(0) + (0)(0) + (0)(0) + (1)(0) = 0$
$d := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (1)(0) = 1$
$e := (1)(0) + (0)(0) + (0)(0) + (0)(1) + (1)(0) = 0$
$f := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(1) = 1$
$g := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 1$
$h := (1)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$i := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 1$
$j := (1)(0) + (0)(0) + (0)(0) + (0)(1) + (0)(0) = 0$
$k := (0)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(1) = 0$
$l := (0)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$m := (0)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$n := (0)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$o := (0)(0) + (0)(0) + (0)(0) + (0)(1) + (0)(0) = 0$
$p := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(1) = 1$
$q := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 1$
$r := (1)(0) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 0$
$s := (1)(1) + (0)(0) + (0)(0) + (0)(0) + (0)(0) = 1$
$t := (1)(0) + (0)(0) + (0)(0) + (0)(1) + (0)(0) = 0$
$u := (0)(1) + (0)(0) + (0)(0) + (1)(0) + (0)(1) = 0$
$v := (0)(1) + (0)(0) + (0)(0) + (1)(0) + (0)(0) = 0$
$w := (0)(0) + (0)(0) + (0)(0) + (1)(0) + (0)(0) = 0$
$x := (0)(1) + (0)(0) + (0)(0) + (1)(0) + (0)(0) = 0$
$y := (0)(0) + (0)(0) + (0)(0) + (1)(1) + (0)(0) = 1$

6 (2a) ***A, B* are as defined on page 202.** Circle the correct choices and record your proofs and witnesses on the facing page. Express your answers as arrow-diagrams. Display the sources and targets as sets with elements.

$\boldsymbol{A} = \{0, 1, 4\}$ $\boldsymbol{\nu}(\boldsymbol{A}) = 3$ $\boldsymbol{B} = \{0, 3, 4\}$ $\boldsymbol{\nu}(\boldsymbol{B}) = 3$

1(i) $\exists \alpha \in Fnc(A,B)\left(\left(Inj(\alpha)\right) \wedge \left(Srj(\alpha)\right)\right)$ (Y) N Pf (W)

1(ii) $\exists \beta \in Fnc(A,B)\left(\left(\cancel{Inj}(\beta)\right) \wedge \left(Srj(\beta)\right)\right)$ Y (N) (Pf) W

1(iii) $\exists \gamma \in Fnc(A,B)\left(\left(Inj(\gamma)\right) \wedge \left(\cancel{Srj}(\gamma)\right)\right)$ Y (N) (Pf) W

1(iv) $\exists \delta \in Fnc(A,B)\left(\left(\cancel{Inj}(\delta)\right) \wedge \left(\cancel{Srj}(\delta)\right)\right)$ (Y) N Pf (W)

1 (v) Find: $Im(\delta) = \{0\}$

1 (vi) **On the facing page**, draw arrow-diagrams for **every adjusted, maximal, bijective restriction** of $\delta: A \longrightarrow B$ and count the total number of **such restrictions**: 3

2a. i

$\alpha := \{4 \to 4,\ 1 \to 3,\ 0 \to 0\} : \{0, 1, 4\} \to \{0, 3, 4\}$

2a. ii

$$\frac{\begin{array}{c}\beta \in Fnc(A,B) \qquad f := Im(\beta) \\ Srj(\beta) \qquad v(A) = 3 = v(B)\end{array}}{\begin{array}{c} v(f) = v(B) = 3 \\ \hline \neg(3 > 3) \\ \hline \neg(v(A) > v(f)) \\ \hline \neg(\neg(Inj(\beta))) \\ \hline Inj(\beta) \\ \hline \nexists\ \beta \in Fnc(A,B)\ \big((Inj(\beta)) \wedge (Srj(\beta))\big)\end{array}}$$

2a. iii

$$\frac{\begin{array}{c}\gamma \in Fnc(A,B) \qquad I := Im \\ Inj(\gamma) \qquad v(A) = 3 = v(B)\end{array}}{\begin{array}{c} v(I) = v(A) = 3 \\ \hline \neg(3 < 3) \\ \hline \neg(v(I) < v(B)) \\ \hline \neg(\neg(Srj(\gamma))) \\ \hline Srj(\gamma) \\ \hline \nexists\ \gamma \in Fnc(A,B)\ \big((Inj(\gamma)) \wedge (Srj(\gamma))\big)\end{array}}$$

2a. iv

$\delta := \{4 \to 4,\ 1 \to 3,\ 0 \to 0\} : \{0, 1, 4\} \to \{0, 3, 4\}$

2a. v. 0

$\delta\big|_{\{0\}} : \{0 \to 0\} : \{0\} \longrightarrow \{0\}$

2a. v. 1

$\delta\big|_{\{1\}} : \{1 \to 0\} : \{1\} \longrightarrow \{0\}$

2a. v. 2

$\delta\big|_{\{4\}} : \{4 \to 0\} : \{4\} \longrightarrow \{0\}$

2a. o

$$\frac{InjFncs(A,B) \neq \{\}}{v(A) \leq v(B)}$$

$$\frac{SrjFncs(A,B) \neq \{\}}{v(A) \geq v(B)}$$

6 (2b) Define the relations $R \subseteq J \times J$ and $S \subseteq E \times E$ as follows:

$$R \coloneqq \left\{ (j\,,k) \in J \times J \;\middle|\; Evn(j+k) \right\} \subseteq J \times J$$

$$S \coloneqq \left\{ (e\,,f) \in E \times E \;\middle|\; \left(e|f\right) \vee \left(f|e\right) \right\} \subseteq E \times E$$

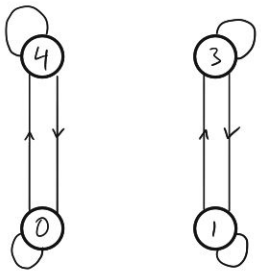
and answer the following questions.

3 (i) Decide if $Eqv(R)$ showing work on the facing side:

(Y) N (Pf) W

If yes, then draw $DG(R)$ on this side.

$DG(R)$:



3 (ii) Decide if $PO(S)$ showing work on the facing side:

Y (N) (Pf) W

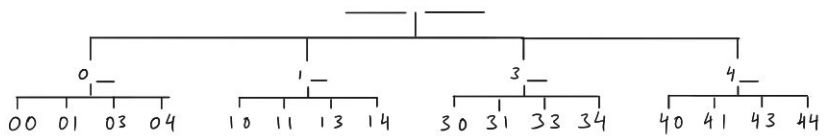
If yes, then draw $HsDgrm(S)$ on this side.

$HsDgrm(S)$:

2b.i.0

Tree

$J \times J$



2b.i.1

$$J \times J = \{0,1,3,4\} \times \{0,1,3,4\} = \begin{Bmatrix} (0,0), (0,1), (0,3), (0,4), (1,0), (1,1), (1,3), (1,4), \\ (3,0), (3,1), (3,3), (3,4), (4,0), (4,1), (4,3), (4,4) \end{Bmatrix}$$

2b.i.2

$$R := \{(j,k) \in J \times J \mid Evn(j+k)\} \subseteq J \times J$$

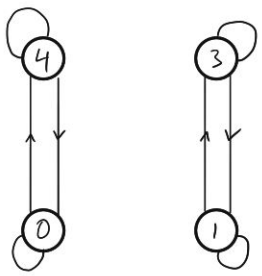
$$\frac{Evn(0)}{\frac{Evn(0+0)}{(0,0) \in R}} \quad \frac{\frac{Odd(1)}{Odd(0+1)}}{\frac{\cancel{Evn}(0+1)}{(0,1) \notin R}} \quad \frac{\cancel{Evn}(0+3)}{(0,3) \notin R} \quad \frac{Evn(0+4)}{(0,4) \in R} \quad \frac{\cancel{Evn}(1+0)}{(1,0) \notin R} \quad \frac{Evn(1+1)}{(1,1) \in R} \quad \frac{Evn(1+3)}{(1,3) \in R} \quad \frac{\cancel{Evn}(1+4)}{(1,4) \notin R}$$

$$\frac{\cancel{Evn}(3+0)}{(3,0) \notin R} \quad \frac{Evn(3+1)}{(3,1) \in R} \quad \frac{Evn(3+3)}{(3,3) \in R} \quad \frac{\cancel{Evn}(3+4)}{(3,4) \notin R} \quad \frac{Evn(4+0)}{(4,0) \in R} \quad \frac{\cancel{Evn}(4+1)}{(4,1) \notin R} \quad \frac{\cancel{Evn}(4+3)}{(4,3) \notin R} \quad \frac{Evn(4+4)}{(4,4) \in R}$$

$$R = \begin{Bmatrix} (0,0), (0,4), (1,1), (1,3), \\ (3,1), (3,3), (4,0), (4,4) \end{Bmatrix}$$

2b.i.3

$$Src(R) = J = Tgt(R)$$



$DG(R)$

$$\frac{Rflx(R) \quad Sym(P) \quad Trns(R)}{Eqv(P)}$$

2b.ii.2

$$S := \{(e,f) \in E \times E \mid (e|f) \vee (f|e)\} \subseteq E \times E$$

$$\frac{(0|0) \vee (0|0)}{(0,0) \in S} \quad \frac{(0|1) \vee (1|0)}{(0,1) \in S} \quad \frac{(0|4) \vee (4|0)}{(0,4) \in S}$$

$$\frac{(1|0) \vee (0|1)}{(1,0) \in S} \quad \frac{(1|1) \vee (1|1)}{(1,1) \in S} \quad \frac{(1|4) \vee (4|1)}{(1,4) \in S}$$

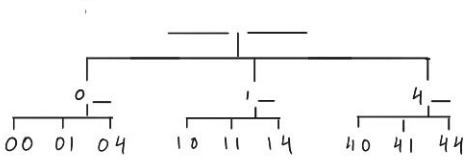
$$\frac{(4|0) \vee (0|4)}{(4,0) \in S} \quad \frac{(4|1) \vee (1|4)}{(4,1) \in S} \quad \frac{(4|4) \vee (4|4)}{(4,4) \in S}$$

$$S = \begin{Bmatrix} (0,0), (0,1), (0,4), \\ (1,0), (1,1), (1,4), \\ (4,0), (4,1) (4,4) \end{Bmatrix}$$

2b.ii.0

Tree

$E \times E$



2.ii.3

$$\frac{\frac{(0,1) \in S}{(1,0) \in S} \quad (1 \neq 0)}{\frac{\neg\left(\forall (a,b) \in \mathcal{E}\left(\frac{(aSb) \wedge (bSa)}{a=b}\right)\right)}{\cancel{AntSym}(S)}}$$

$$\cancel{PO}(S)$$

2b.ii.1

$$E \times E = \{0,1,4\} \times \{0,1,4\} = \begin{Bmatrix} (0,0), (0,1), (0,4), \\ (1,0), (1,1), (1,4), \\ (4,0), (4,1) (4,4) \end{Bmatrix}$$

2210

6 (3a) Define the relations $P \subseteq J \times J$ and $Q \subseteq E \times E$ as follows:

$$P \coloneqq \left\{ (j\,,k) \in J \times J \;\middle|\; j|k \right\} \quad \subseteq \quad J \times J$$

$$Q \coloneqq \left\{ (e\,,f) \in E \times E \;\middle|\; f \mid e \right\} \quad \subseteq \quad E \times E$$

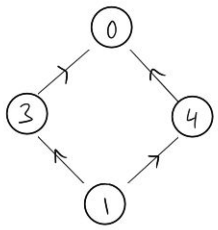
and answer the following questions.

3 (i) Decide if $PO(P)$ showing work on the facing side:

(Y) N (Pf) W

If yes, then draw $HsDgrm(P)$ on this side.

$HsDgrm(P)$:



3 (ii) Decide if $PO(Q)$ showing work on the facing side:

(Y) N (Pf) W

If yes, then draw $HsDgrm(Q)$ on this side.

$HsDgrm(Q)$:



3a.i.0

$J \times J$ TREE done on pg. 2210

$= \{0,1,3,4\} \times \{0,1,3,4\}$

$= \begin{Bmatrix} (0,0), (0,1), (0,3), (0,4), (1,0), (1,1), (1,3), (1,4), \\ (3,0), (3,1), (3,3), (3,4), (4,0), (4,1), (4,3), (4,4) \end{Bmatrix}$

3a.i.1

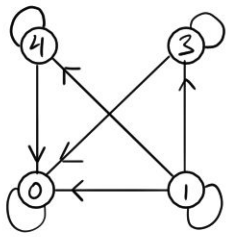
$$P := \{(j,k) \in J \times J \mid j \mid k\} \subseteq J \times J$$

$0 \mid 0$	$0 \nmid 1$	$0 \nmid 3$	$0 \nmid 4$	$1 \mid 0$	$1 \mid 1$	$1 \mid 3$	$1 \mid 4$
$(0,0) \in P$	$(0,1) \notin P$	$(0,3) \notin P$	$(0,4) \notin P$	$(1,0) \in P$	$(1,1) \in P$	$(1,3) \in P$	$(1,4) \in P$
$3 \mid 0$	$3 \nmid 1$	$3 \mid 3$	$3 \nmid 4$	$4 \mid 0$	$4 \nmid 1$	$4 \nmid 3$	$4 \mid 4$
$(3,0) \in P$	$(3,1) \notin P$	$(3,3) \in P$	$(3,4) \notin P$	$(4,0) \in P$	$(4,1) \notin P$	$(4,3) \notin P$	$(4,4) \in P$

$$P = \begin{Bmatrix} (0,0), & (1,0), & (1,1), & (1,3), & (1,4), \\ (3,0), & (3,3), & (4,0), & (4,4) & \end{Bmatrix}$$

3a.i.2

$$Src(P) = J = Tgt(P)$$



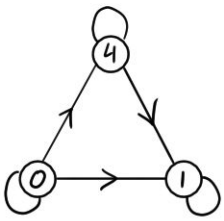
$DG(P)$

$Rflx(P)$ $AntSym(P)$ $Trns(P)$

$PO(P)$

3a.ii.2

$$Src(Q) = E = Tgt(Q)$$



$DG(Q)$

$Rflx(Q)$ $AntSym(Q)$ $Trns(Q)$

$PO(Q)$

3a.ii.0

TREE done on pg. 2210

$E \times E$

$= \{0,1,4\} \times \{0,1,4\}$

$= \begin{Bmatrix} (0,0), (0,1), (0,4), \\ (1,0), (1,1), (1,4), \\ (4,0), (4,1), (4,4) \end{Bmatrix}$

3a.ii.1

$$Q := \{(e,f) \in E \times E \mid (f \mid e)\} \subseteq E \times E$$

$0 \mid 0$	$1 \mid 0$	$4 \mid 0$
$(0,0) \in Q$	$(0,1) \in Q$	$(0,4) \in Q$
$0 \nmid 1$	$1 \mid 1$	$4 \nmid 1$
$(1,0) \notin Q$	$(1,1) \in Q$	$(1,4) \notin Q$
$0 \nmid 4$	$1 \mid 4$	$4 \mid 4$
$(4,0) \notin Q$	$(4,1) \in Q$	$(4,4) \in Q$

$$Q = \{(0,0), (0,1), (1,1), (4,1), (0,4), (4,4)\}$$

6 (3b) Given a set S and two equivalence relations $P, Q \in Rln\left(S, S\right)$

circle the correct choices in the below and **provide an answer** accordingly **on the facing side**:

2 (i) $Eqv\left(P^{op}\right)$ (Y) N (Pf) W

3 (ii) $Eqv\left(P \circ Q\right)$ Y (N) Pf (W)

3b.i 1

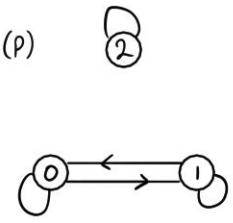
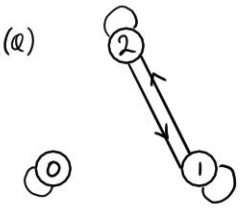
$$\frac{Eqv(P)}{Sym(P)} \qquad \frac{P = P^{op}}{Eqv(P^{op})}$$

3b.ii.0

$S := (0..2) \in \mathcal{U}$

$Q := \{(0,0), (1,1), (2,2), (1,2), (2,1)\} \subseteq S \times S$

$P := \{(0,0), (1,1), (2,2), (0,1), (1,0)\} \subseteq S \times S$

$DG(P)$ 	$DG(Q)$ 
$Rflx(P)$ $Sym(P)$ $Trns(P)$	$Rflx(Q)$ $Sym(Q)$ $Trns(Q)$
$Eqv(P)$	$Eqv(Q)$

$P \circ Q = \{(0,0), (1,1), (2,2), (0,1), (1,0), (1,2), (2,1), (2,0)\}$

$\cancel{Eqv}(P \circ Q)$

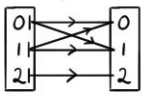
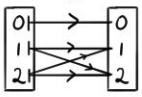
$\exists S \in \mathcal{U} \left(\exists P, Q \in EqvRlns(S) \left(\cancel{Eqv}(P \circ Q) \right) \right)$

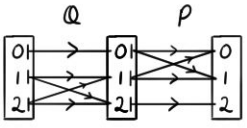
$\neg \left(\begin{array}{l} \forall S \in \mathcal{U} \\ \forall P, Q \in EqvRlns(S) \end{array} \left(Eqv(P \circ Q) \right) \right)$

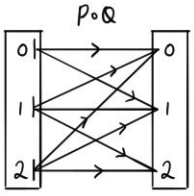
3b.ii.1

$P := \{(0,0), (1,1), (2,2), (0,1), (1,0)\}$

$Q := \{(0,0), (1,1), (2,2), (1,2), (2,1)\}$

P 	Q 
$AD(P)$	$AD(Q)$

$AD(P \circ Q)$ 

$AD(P \circ Q)$ 

$P \circ Q = \{(0,0), (1,1), (2,2), (0,1), (1,0), (1,2), (2,1), (2,0)\}$

$(2,0) \in (P \circ Q) \qquad (0,2) \notin (P \circ Q)$

$\cancel{Eqv}(P \circ Q)$